

The Relational Nature of Information

Why Information Is Neither Matter nor Energy

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Abstract

Information is commonly treated either as a measurable quantity (Shannon), a semantic entity (Floridi), or a physical resource ("It from Bit"). This paper proposes a different ontological perspective: information is not an intrinsic property of matter, nor of structure alone, but an emergent property of a relation established between systems through a shared structural code.

The distinction naturally separates four conceptual layers—matter, structure, coding, and information—and explains why identical physical configurations may carry information for one system while remaining meaningless for another. This relational framework unifies examples from telecommunications, genetics, language, computation, and consciousness without reducing information to either physics or semantics.

1. Introduction

What is information?

Despite decades of research, no universally accepted ontological definition exists.

Three major traditions dominate contemporary thought.

Shannon identifies information with statistical uncertainty.

Physics increasingly treats information as a conserved physical quantity.

Philosophy often defines information through meaning or semantic content.

Each perspective captures an important aspect of information.

Yet each leaves unanswered a fundamental question:

Where does information actually exist?

This paper argues that information does not exist as an intrinsic property of objects.

Instead, it emerges from structural relations between systems.

2. The Problem

Consider a modem transmitting a signal.

The wire contains electrons.

The signal possesses amplitude, frequency and phase.

Physics completely describes these quantities.

But nowhere inside Maxwell's equations appears the binary sequence

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101101
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The bits appear only because transmitter and receiver share a coding protocol.

Without that protocol,

the same electromagnetic wave becomes merely a physical oscillation.

The physical structure remains identical.

The information disappears.

3. Four Ontological Layers

The confusion surrounding information arises because four different categories are often conflated.

Layer 1 — Matter

Matter provides the physical substrate.

Examples:

- electrons
- photons
- molecules
- neurons

Matter exists independently of interpretation.

Layer 2 — Structure

Structure organizes matter.

Examples:

- waveforms
- molecular geometry
- neural connectivity
- DNA sequences

Structure also exists objectively.

Layer 3 — Coding

Coding establishes correspondences between structures.

Examples:

Phase = 180°

↓

Binary 1

or

Codon AUG

↓

Methionine

The coding rule is neither matter nor structure.

It is a relational mapping.

Layer 4 — Information

Information appears only when a structure is interpreted through a shared code.

Therefore,

Information

≠

Matter

Information

≠

Structure

Information

=

Structure

•

Code

•

4. The Relational Information Principle

We propose:

Relational Information Principle

Information is not an intrinsic property of physical systems.

It is an emergent property of relations established through shared structural codes.

Symbolically,

$$Information = R(S, C)$$

where

S

is a physical structure

and

C

is a coding relation.

Without

C ,

only structure exists.

5. Examples

Telecommunications

A modem signal carries information only because transmitter and receiver negotiated the same protocol.

Without synchronization,

only electromagnetic waves exist.

DNA

DNA contains nucleotides.

But nucleotides become genetic information only because cellular machinery implements the genetic code.

Without ribosomes,

DNA becomes chemistry.

Human Language

A Chinese book contains ink patterns.

For someone unable to read Chinese,

those patterns remain physical structures.

Information appears only for readers sharing the linguistic code.

Music

A musical score is paper and ink.

Only musicians sharing musical notation recover the music.

Computing

Machine code consists of voltage levels.

Processors interpret those voltages according to an instruction set architecture.

Without that architecture,

there is no software.

6. Consequences

The relational view explains why

identical physical structures

may simultaneously

contain information

and

contain none,

depending upon the relational framework.

Information therefore cannot be reduced either to matter or to geometry alone.

7. Consciousness

The same distinction illuminates consciousness.

Neural activity constitutes physical structure.

Experience, however, cannot be identified with the physical description alone.

A complete neural map remains a description.

It is not identical to lived experience.

The relational framework therefore distinguishes

description,

structure,

and realization,

without requiring reduction between them.

8. Discussion

This proposal differs from existing approaches.

Shannon measures uncertainty.

Physics studies state evolution.

Semantic theories analyze meaning.

The present proposal addresses a different question:

the ontology of information itself.

Information is neither an object nor a substance.

It is a relational property emerging whenever structural configurations become mutually interpretable.

9. Conclusion

Matter carries physical states.

Structure organizes matter.

Coding establishes correspondences.

Information emerges only through relations between systems capable of sharing those correspondences.

The implication is significant.

Information is not "stored" inside matter.

Rather,

matter supports structures,

structures enable coding,

and coding allows information to emerge.

Information is therefore best understood not as a thing,

but as a relation.

Appendix A – Formalization of the Relational Information Principle

The relational framework presented in this paper is intentionally independent of the formal machinery developed in the Theory of Axiomatic Necessity (TNA). Nevertheless, both approaches are structurally compatible.

The present work identifies four conceptual layers that precede the emergence of information:

RELATIONAL INFORMATION FRAMEWORK	TNA CORRESPONDENCE
Matter	Physical substrate (pre-formal domain)
Structure	$P(\Omega)$: projected structural configurations
Coding	R : Coherence Restriction
Interpretation	I : Instantiation Operator
Information	Relational realization within N_1

The correspondence should not be interpreted as an identity between concepts but as a structural mapping between two complementary descriptive levels.

In the TNA framework, projected structures $P(\Omega)$ represent admissible structural configurations independently of their realization. The Coherence Restriction R determines which projected configurations satisfy the structural constraints required for admissibility.

The present paper interprets this restriction as the existence of a shared structural code between systems.

Instantiation occurs through the operator I , which realizes an admissible structure within the domain N_1 .

Consequently, information is not identified with the projected structure itself, but with the relational realization established through successful instantiation.

This correspondence can be summarized schematically as

$$\text{Matter} \longrightarrow P(\Omega) \xrightarrow{R} I \longrightarrow N_1$$

where the emergence of information requires every preceding stage.

Relational Information Constraint

The relational interpretation proposed in this paper can be expressed formally.

Let

- S denote a physical structure,
- C a shared coding relation,
- I_R the resulting relational information.

Then

$$I_R = \Phi(S, C)$$

where

$$\Phi(S, \emptyset) = \emptyset.$$

This expresses the central claim of the paper:

A physical structure alone does not constitute information. Information emerges only when a structure participates in a shared coding relation.

Accordingly, no universal mapping

$$g : S \rightarrow I_R$$

exists independently of the coding relation.

Information is therefore **non-intrinsic**: it is an emergent property of structured relations rather than an inherent property of physical objects.

Compatibility with TNA

The present framework is fully compatible with the TNA Principle of Non-Derivability.

Projected structures may exist independently of realization, but realization cannot be recovered from structural description alone.

Likewise, physical structures may exist independently of information, but information cannot emerge without the relational coding required for interpretation.

Both theories therefore distinguish between **structural existence** and **relational realization**, although they address different aspects of that distinction.

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